Functions of Several Variables & Partial Derivatives

Evaluating functions of two variables

1 For $f(x, y) = x^2y - 3xy + y^2$, find $f(2, -1)$.

2 For $f(x, y) = \sqrt{x^2 + y^2}$, find $f(3, 4)$ and describe the domain of f.

3 For $g(x, y, z) = xy + yz + zx$, find $g(1, 2, 3)$.

4 For $f(x, y) = \dfrac{x - y}{x + y}$, find $f(5, 3)$ and state one point where f is undefined.

Computing partial derivatives

5 For $f(x, y) = x^4y^3 + 2xy - y^5$, find f_x and f_y.

6 For $f(x, y) = e^{x^2y}$, find f_x and f_y.

7 For $f(x, y) = \ln(x^2 + y^2)$, find f_x and f_y.

8 For $f(x, y) = x\sin(xy)$, find f_x and f_y.

Second-order and mixed partial derivatives

9 For $f(x, y) = x^3y^2$, find f_{xx}, f_{yy}, and f_{xy}.

10 For $f(x, y) = e^{xy}$, find f_{xx} and f_{xy}.

11 For $f(x, y) = \sin(x)\cos(y)$, find f_{xy}.

Clairaut's Theorem: equality of mixed partials

12 For $f(x, y) = \sin(xy) + x^2y$, verify that $f_{xy} = f_{yx}$.

13 State Clairaut's Theorem and the condition it requires.

14 For $f(x, y) = x^3y^4$, compute f_{xy} and f_{yx} separately to confirm Clairaut's Theorem.

Domains and level curves

15 Find the domain of $f(x, y) = \ln(x - y)$.

16 Describe the level curve $f(x, y) = 4$ for $f(x, y) = x^2 + y^2$.

17 Find the domain of $f(x, y) = \sqrt{9 - x^2 - y^2}$ and describe it geometrically.

18 Describe the level curves $f(x, y) = c$ for $f(x, y) = y - x^2$, for a general constant c.

Functions of three variables

19 For $f(x, y, z) = x^2 + y^2 + z^2$, find f_x, f_y, f_z.

20 Describe the level surface $f(x, y, z) = 9$ for $f(x, y, z) = x^2 + y^2 + z^2$.

21 For $w = x^2yz^3$, find w_x, w_y, and w_z.

Interpreting partial derivatives

22 The temperature at a point on a metal plate is $T(x, y) = 50 - 0.1x^2 - 0.2y^2$. Compute $T_x(2, 3)$ and interpret its sign.

23 For the production function $P(L, K) = 50L^{0.6}K^{0.4}$ (Cobb–Douglas, L=labor, K=capital), state in words what P_L represents.

Higher-order partial derivatives with three variables

24 For $f(x, y, z) = x^2yz + y^3z^2$, find f_{xz}.

25 For $f(x, y, z) = e^{xyz}$, find f_x, then f_{xy}.

Sketching and identifying surfaces from level curves

26 The level curves of $f(x, y) = x^2 - y^2$ at $c = 1, -1, 0$ are hyperbolas (and a pair of crossing lines at $c = 0$). Find the equations of the level curves for $c = 4$ and $c = -4$.

27 For $f(x, y) = \dfrac{1}{x^2 + y^2 + 1}$, describe the level curve $f(x, y) = \dfrac{1}{2}$.

28 Explain why the level curves of any function $f(x, y) = g(x^2 + y^2)$, for a one-variable function g with $g' \neq 0$ everywhere, are always circles centered at the origin.